



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 23, 24: Advanced Topic*

Yajing Shen (申亞京)

Office: 2450

eeyajing@ust.hk

Frequency-Domain vs. State-Space

- ▶ 90% of industrial controllers are designed using frequency-domain methods (PID is a popular architecture)
- ▶ 90% of current research in systems and control is in the state-space framework

Frequency-Domain vs. State-Space

- ▶ 90% of industrial controllers are designed using frequency-domain methods (PID is a popular architecture)
- ▶ 90% of current research in systems and control is in the state-space framework

To be able to talk to control engineers *and* follow progress in the field, we need to know both methods and *understand the connections between them*.

A General State-Space Model

$$\begin{aligned} \text{state } x &= \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n & \text{input } u &= \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix} \in \mathbb{R}^m \\ \text{output } y &= \begin{pmatrix} y_1 \\ \vdots \\ y_p \end{pmatrix} \in \mathbb{R}^p \end{aligned}$$

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

where:

A – system matrix ($n \times n$)

B – input matrix ($n \times m$)

C – output matrix ($p \times n$)

D – feedthrough matrix ($p \times m$)

From State-Space to Transfer Function

The transfer function from u to y , corresponding to

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

is given by

$$G(s) = C(Is - A)^{-1}B + D$$

Observe that $G(s)$ contains information about the state-space matrices A, B, C, D !!

From State-Space to Transfer Function

$$\begin{aligned}\dot{x} &= Ax + Bu & Y(s) &= G(s)U(s) \\ y &= Cx + Du & &= [C(Is - A)^{-1}B + D] U(s)\end{aligned}$$

Important!!

- ▶ $G(s)$ is *undefined* when the $n \times n$ matrix $Is - A$ is *singular* (or noninvertible), i.e., precisely when $\det(Is - A) = 0$
- ▶ since A is $n \times n$, $\det(Is - A)$ is a *polynomial* of degree n (the *characteristic polynomial* of A):

$$\det(Is - A) = \det \begin{pmatrix} s - a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & s - a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \dots & s - a_{nn} \end{pmatrix},$$

and its roots are the *eigenvalues* of A

From State-Space to Transfer Function

$$\begin{aligned} \dot{x} &= Ax + Bu & Y(s) &= G(s)U(s) \\ y &= Cx + Du & &= [C(Is - A)^{-1}B + D] U(s) \end{aligned}$$

Important!!

- ▶ $G(s)$ is *undefined* when the $n \times n$ matrix $Is - A$ is *singular* (or noninvertible), i.e., precisely when $\det(Is - A) = 0$
- ▶ since A is $n \times n$, $\det(Is - A)$ is a *polynomial* of degree n (the *characteristic polynomial* of A):

$$\det(Is - A) = \det \begin{pmatrix} s - a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & s - a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \dots & s - a_{nn} \end{pmatrix},$$

and its roots are the *eigenvalues* of A

- ▶ G is (open-loop) stable if all eigenvalues of A lie in LHP.

Example: Computing $G(s)$

Consider the state-space model in **Controller Canonical Form (CCF)***:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} 1 & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

— this is a *single-input, single-output* (SISO) system, since $u, y \in \mathbb{R}$; the state is two-dimensional.

Let's compute the transfer function:

$$G(s) = C(Is - A)^{-1}B \quad (D = 0 \text{ here})$$

$$Is - A = \begin{pmatrix} s & -1 \\ 6 & s + 5 \end{pmatrix}$$

* We will explain this terminology later.

Example: Computing $G(s)$

$$Is - A = \begin{pmatrix} s & -1 \\ 6 & s + 5 \end{pmatrix} \quad \text{— how do we compute } (Is - A)^{-1}?$$

Example: Computing $G(s)$

$$Is - A = \begin{pmatrix} s & -1 \\ 6 & s + 5 \end{pmatrix} \quad \text{— how do we compute } (Is - A)^{-1}?$$

A useful formula for the inverse of a 2×2 matrix:

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \det M \neq 0 \implies M^{-1} = \frac{1}{\det M} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Applying the formula, we get

$$\begin{aligned} (Is - A)^{-1} &= \frac{1}{\det(Is - A)} \begin{pmatrix} s + 5 & 1 \\ -6 & s \end{pmatrix} \\ &= \frac{1}{s^2 + 5s + 6} \begin{pmatrix} s + 5 & 1 \\ -6 & s \end{pmatrix} \end{aligned}$$

Example: Computing $G(s)$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{(1 \quad 1)}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\begin{aligned} G(s) &= C(Is - A)^{-1}B \\ &= (1 \quad 1) \frac{1}{s^2 + 5s + 6} \begin{pmatrix} s + 5 & 1 \\ -6 & s \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \frac{1}{s^2 + 5s + 6} (1 \quad 1) \begin{pmatrix} 1 \\ s \end{pmatrix} \\ &= \frac{s + 1}{s^2 + 5s + 6} \end{aligned}$$

- the above state-space model is a *realization* of this t.f.

Example: Computing $G(s)$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{(1 \quad 1)}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\begin{aligned} G(s) &= C(Is - A)^{-1}B \\ &= (1 \quad 1) \frac{1}{s^2 + 5s + 6} \begin{pmatrix} s + 5 & 1 \\ -6 & s \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \frac{1}{s^2 + 5s + 6} (1 \quad 1) \begin{pmatrix} 1 \\ s \end{pmatrix} \\ &= \frac{s + 1}{s^2 + 5s + 6} \end{aligned}$$

- ▶ the above state-space model is a *realization* of this t.f.
- ▶ note how coefficients 5 and 6 appear in both $G(s)$ and A !!

State-Space Realizations of Transfer Functions

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -\textcolor{red}{6} & -\textcolor{blue}{5} \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{(1 \quad 1)}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$
$$G(s) = \frac{s + 1}{s^2 + \textcolor{blue}{5}s + \textcolor{red}{6}}$$

— at least in this example, information about the state-space model (A, B, C) is contained in $G(s)$.

State-Space Realizations of Transfer Functions

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -\textcolor{red}{6} & -\textcolor{blue}{5} \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{(1 \quad 1)}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$
$$G(s) = \frac{s + 1}{s^2 + \textcolor{blue}{5}s + \textcolor{red}{6}}$$

— at least in this example, information about the state-space model (A, B, C) is contained in $G(s)$.

Is this information *recoverable*? — i.e., is there only one state-space realization of a given t.f.? Or are there many?

State-Space Realizations of Transfer Functions

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} 1 & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$
$$G(s) = \frac{s + 1}{s^2 + 5s + 6}$$

— at least in this example, information about the state-space model (A, B, C) is contained in $G(s)$.

Is this information *recoverable*? — i.e., is there only one state-space realization of a given t.f.? Or are there many?

Answer: There are infinitely many!

State-Space Realizations of Transfer Functions

Start with

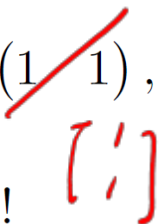
$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} 1 & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

and consider a new state-space model

$$\dot{x} = \bar{A}x + \bar{B}u, \quad y = \bar{C}x$$

with

$$\bar{A} = A^T = \begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix}, \quad \bar{B} = C^T = \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad \bar{C} = B^T = \begin{pmatrix} 0 & 1 \end{pmatrix}$$



This is a different state-space model!

State-Space Realizations of Transfer Functions

Claim: The state-space model

$$\dot{x} = \bar{A}x + \bar{B}u, \quad y = \bar{C}x$$

with

$$\bar{A} = A^T, \quad \bar{B} = C^T, \quad \bar{C} = B^T$$

has the same transfer function as the original model with (A, B, C) .

State-Space Realizations of Transfer Functions

Claim: The state-space model

$$\dot{x} = \bar{A}x + \bar{B}u, \quad y = \bar{C}x$$

with

$$\bar{A} = A^T, \quad \bar{B} = C^T, \quad \bar{C} = B^T$$

has the same transfer function as the original model with (A, B, C) .

Proof:

$$\begin{aligned} \bar{C}(Is - \bar{A})^{-1}\bar{B} &= B^T (Is - A^T)^{-1} C^T \\ &= B^T [(Is - A)^T]^{-1} C^T \\ &= B^T \left[(Is - A)^{-1} \right]^T C^T \\ &= [C(Is - A)^{-1}B]^T \\ &= C(Is - A)^{-1}B \end{aligned}$$

State-Space Realizations of Transfer Functions

The state-space model

$$\dot{x} = \bar{A}x + \bar{B}u, \quad y = \bar{C}x$$

with

$$\bar{A} = A^T, \quad \bar{B} = C^T, \quad \bar{C} = B^T$$

has the same transfer function as the original model with (A, B, C) .

But the state-space model is now in the **Observer Canonical Form (OCF)**:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} u, \quad y = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

State-Space Realizations: Bottom Line

- ▶ a given transfer function $G(s)$ can be realized using infinitely many state-space models
- ▶ certain properties make some realizations preferable to others
- ▶ one such property is *controllability*

Controllability Matrix

Consider a single-input system ($u \in \mathbb{R}$):

$$\dot{x} = Ax + Bu, \quad y = Cx \quad x \in \mathbb{R}^n$$

The **Controllability Matrix** is defined as

$$\mathcal{C}(A, B) = [B \mid AB \mid A^2B \mid \dots \mid A^{n-1}B]$$

- recall that A is $n \times n$ and B is $n \times 1$, so $\mathcal{C}(A, B)$ is $n \times n$;
- the controllability matrix only involves A and B , not C

We say that the above system is **controllable** if its controllability matrix $\mathcal{C}(A, B)$ is *invertible*.

(This definition is only true for the single-input case; the multiple-input case involves the *rank* of $\mathcal{C}(A, B)$.)

Controllability Matrix

Consider a single-input system ($u \in \mathbb{R}$):

$$\dot{x} = Ax + Bu, \quad y = Cx \quad x \in \mathbb{R}^n$$

The **Controllability Matrix** is defined as

$$\mathcal{C}(A, B) = [B \mid AB \mid A^2B \mid \dots \mid A^{n-1}B]$$

We say that the above system is **controllable** if its controllability matrix $\mathcal{C}(A, B)$ is *invertible*.

- ▶ As we will see later, if the system is controllable, then we may assign arbitrary closed-loop poles by *state feedback* of the form $u = -Kx$.
- ▶ Whether or not the system is controllable depends on its state-space realization.

Example: Computing $\mathcal{C}(A, B)$

Let's get back to our old friend:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} 1 & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Here, $x \in \mathbb{R}^2 \implies A \in \mathbb{R}^{2 \times 2} \implies \mathcal{C}(A, B) \in \mathbb{R}^{2 \times 2}$

Example: Computing $\mathcal{C}(A, B)$

Let's get back to our old friend:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} 1 & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Here, $x \in \mathbb{R}^2 \implies A \in \mathbb{R}^{2 \times 2} \implies \mathcal{C}(A, B) \in \mathbb{R}^{2 \times 2}$

$$\begin{aligned} \mathcal{C}(A, B) &= [B \mid AB] & AB &= \begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \end{pmatrix} \\ \implies \mathcal{C}(A, B) &= \begin{pmatrix} 0 & 1 \\ 1 & -5 \end{pmatrix} \end{aligned}$$

Is this system controllable?

Example: Computing $\mathcal{C}(A, B)$

Let's get back to our old friend:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} 1 & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Here, $x \in \mathbb{R}^2 \implies A \in \mathbb{R}^{2 \times 2} \implies \mathcal{C}(A, B) \in \mathbb{R}^{2 \times 2}$

$$\begin{aligned} \mathcal{C}(A, B) &= [B \mid AB] & AB &= \begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \end{pmatrix} \\ \implies \mathcal{C}(A, B) &= \begin{pmatrix} 0 & 1 \\ 1 & -5 \end{pmatrix} \end{aligned}$$

Is this system controllable?

$$\det \mathcal{C} = -1 \neq 0 \quad \implies \quad \text{system is controllable}$$

Controller Canonical Form

A single-input state-space model

$$\dot{x} = Ax + Bu, \quad y = Cx$$

is said to be in **Controller Canonical Form** (CCF) if the matrices A, B are of the form

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ * & * & * & \dots & * & * \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

A system in CCF is *always controllable!!*

(The proof of this for $n > 2$ uses the Jordan canonical form, we will not worry about this.)

CCF with Arbitrary Zeros

In our example, we had $G(s) = \frac{s+1}{s^2+5s+6}$, with a minimum-phase zero at $z = -1$.

Let's consider a general zero location $s = z$:

$$G(s) = \frac{s - z}{s^2 + 5s + 6}$$

This gives us a CCF realization

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} -z & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Since A, B are the same, $\mathcal{C}(A, B)$ is the same $\implies$ the system is still controllable.

CCF with Arbitrary Zeros

In our example, we had $G(s) = \frac{s+1}{s^2+5s+6}$, with a minimum-phase zero at $z = -1$.

Let's consider a general zero location $s = z$:

$$G(s) = \frac{s - z}{s^2 + 5s + 6}$$

This gives us a CCF realization

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} -z & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Since A, B are the same, $\mathcal{C}(A, B)$ is the same $\implies$ the system is still controllable.

A system in CCF is controllable for any locations of the zeros.

OCF with Arbitrary Zeros

Start with the CCF

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} -z & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

OCF with Arbitrary Zeros

Start with the CCF

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} -z & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Convert to OCF: $(A \mapsto A^T, B \mapsto C^T, C \mapsto B^T)$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix}}_{\bar{A}=A^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} -z \\ 1 \end{pmatrix}}_{\bar{B}=C^T} u, \quad y = \underbrace{\begin{pmatrix} 0 & 1 \end{pmatrix}}_{\bar{C}=B^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

OCF with Arbitrary Zeros

Start with the CCF

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B u, \quad y = \underbrace{\begin{pmatrix} -z & 1 \end{pmatrix}}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Convert to OCF: $(A \mapsto A^T, B \mapsto C^T, C \mapsto B^T)$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix}}_{\bar{A}=A^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} -z \\ 1 \end{pmatrix}}_{\bar{B}=C^T} u, \quad y = \underbrace{\begin{pmatrix} 0 & 1 \end{pmatrix}}_{\bar{C}=B^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

We already know that this system realizes the same t.f. as the original system.

But is it *controllable*?

OCF with Arbitrary Zeros

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix}}_{\bar{A}=A^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} -z \\ 1 \end{pmatrix}}_{\bar{B}=C^T} u, \quad y = \underbrace{\begin{pmatrix} 0 & 1 \end{pmatrix}}_{\bar{C}=B^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Let's find the controllability matrix:

OCF with Arbitrary Zeros

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix}}_{\bar{A}=A^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} -z \\ 1 \end{pmatrix}}_{\bar{B}=C^T} u, \quad y = \underbrace{\begin{pmatrix} 0 & 1 \end{pmatrix}}_{\bar{C}=B^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Let's find the controllability matrix:

$$\mathcal{C}(\bar{A}, \bar{B}) = [\bar{B} \mid \bar{A}\bar{B}] \quad \bar{A}\bar{B} = \begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix} \begin{pmatrix} -z \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ -z-5 \end{pmatrix}$$

$$\therefore \mathcal{C}(\bar{A}, \bar{B}) = \begin{pmatrix} -z & -6 \\ 1 & -z-5 \end{pmatrix}$$

$$\det \mathcal{C} = z(z+5) + 6 = z^2 + 5z + 6 = 0 \quad \text{for } z = -2 \text{ or } z = -3$$

OCF with Arbitrary Zeros

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix}}_{\bar{A}=A^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} -z \\ 1 \end{pmatrix}}_{\bar{B}=C^T} u, \quad y = \underbrace{\begin{pmatrix} 0 & 1 \end{pmatrix}}_{\bar{C}=B^T} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Let's find the controllability matrix:

$$\mathcal{C}(\bar{A}, \bar{B}) = [\bar{B} \mid \bar{A}\bar{B}] \quad \bar{A}\bar{B} = \begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix} \begin{pmatrix} -z \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ -z-5 \end{pmatrix}$$

$$\therefore \mathcal{C}(\bar{A}, \bar{B}) = \begin{pmatrix} -z & -6 \\ 1 & -z-5 \end{pmatrix}$$

$$\det \mathcal{C} = z(z+5) + 6 = z^2 + 5z + 6 = 0 \quad \text{for } z = -2 \text{ or } z = -3$$

The OCF realization of the transfer function

$G(s) = \frac{s-z}{s^2+5s+6}$ is not controllable when $z = -2$ or -3 , even though the CCF is always controllable.

Beware of Pole-Zero Cancellations!

The OCF realization of the transfer function

$$G(s) = \frac{s - z}{s^2 + 5s + 6}$$

is not controllable when $z = -2$ or -3 , even though the CCF is always controllable.

Beware of Pole-Zero Cancellations!

The OCF realization of the transfer function

$$G(s) = \frac{s - z}{s^2 + 5s + 6}$$

is not controllable when $z = -2$ or -3 , even though the CCF is always controllable.

Let's examine $G(s)$ when $z = -2$:

$$G(s) = \frac{s - z}{s^2 + 5s + 6} \Big|_{z=-2} = \frac{\cancel{s+2}}{(\cancel{s+2})(s+3)} = \frac{1}{s+3}$$

— pole-zero cancellation!

For $z = -2$, $G(s)$ is a first-order transfer function, which can always be realized by this 1st-order controllable model:

$$\dot{x}_1 = -3x_1 + u, \quad y = x_1 \quad \longrightarrow \quad G(s) = \frac{1}{s+3}$$

Beware of Pole-Zero Cancellations!!

We can look at this from another angle: consider the t.f.

$$G(s) = \frac{1}{s + 3}$$

We can realize it using a one-dimensional controllable state-space model

$$\dot{x}_1 = -3x_1 + u, \quad y = x_1$$

Beware of Pole-Zero Cancellations!!

We can look at this from another angle: consider the t.f.

$$G(s) = \frac{1}{s+3}$$

We can realize it using a one-dimensional controllable state-space model

$$\dot{x}_1 = -3x_1 + u, \quad y = x_1$$

or a noncontrollable two-dimensional state-space model

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} u, \quad y = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

— certainly not the best way to realize a simple t.f.!

Beware of Pole-Zero Cancellations!!

We can look at this from another angle: consider the t.f.

$$G(s) = \frac{1}{s+3}$$

We can realize it using a one-dimensional controllable state-space model

$$\dot{x}_1 = -3x_1 + u, \quad y = x_1$$

or a noncontrollable two-dimensional state-space model

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} u, \quad y = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

— certainly not the best way to realize a simple t.f.!

Thus, even the *state dimension* of a realization of a given t.f. is not unique!!

Beware of Pole-Zero Cancellations!!

Here is a **really bad** realization of the t.f.

$$G(s) = \frac{1}{s + 3}.$$

Use a two-dimensional model:

$$\dot{x}_1 = -3x_1 + u$$

$$\dot{x}_2 = 100x_2$$

$$y = x_1$$

- ▶ x_2 is not affected by the input u (i.e., it is an uncontrollable mode), and not visible from the output y
- ▶ does not change the transfer function
- ▶ ... and yet, horrible to implement: $x_2(t) \propto e^{100t}$

The transfer function can mask undesirable internal state behavior!!

Pole-Zero Cancellations and Stability

- ▶ In case of a pole-zero cancellation, the t.f. contains *much less* information than the state-space model because some dynamics are “hidden.”
- ▶ These dynamics can be either good (stable) or bad (unstable), but we cannot tell from the t.f.
- ▶ Our original definition of stability (no RHP poles) is flawed because there can be RHP eigenvalues of the system matrix A that are canceled by zeros, yet they still have dynamics associated with them.

Pole-Zero Cancellations and Stability

- ▶ In case of a pole-zero cancellation, the t.f. contains *much less* information than the state-space model because some dynamics are “hidden.”
- ▶ These dynamics can be either good (stable) or bad (unstable), but we cannot tell from the t.f.
- ▶ Our original definition of stability (no RHP poles) is flawed because there can be RHP eigenvalues of the system matrix A that are canceled by zeros, yet they still have dynamics associated with them.

Definition of Internal Stability (State-Space Version): a state-space model with matrices (A, B, C, D) is *internally stable* if all eigenvalues of the A matrix are in LHP.

Pole-Zero Cancellations and Stability

- ▶ In case of a pole-zero cancellation, the t.f. contains *much less* information than the state-space model because some dynamics are “hidden.”
- ▶ These dynamics can be either good (stable) or bad (unstable), but we cannot tell from the t.f.
- ▶ Our original definition of stability (no RHP poles) is flawed because there can be RHP eigenvalues of the system matrix A that are canceled by zeros, yet they still have dynamics associated with them.

Definition of Internal Stability (State-Space Version): a state-space model with matrices (A, B, C, D) is *internally stable* if all eigenvalues of the A matrix are in LHP.

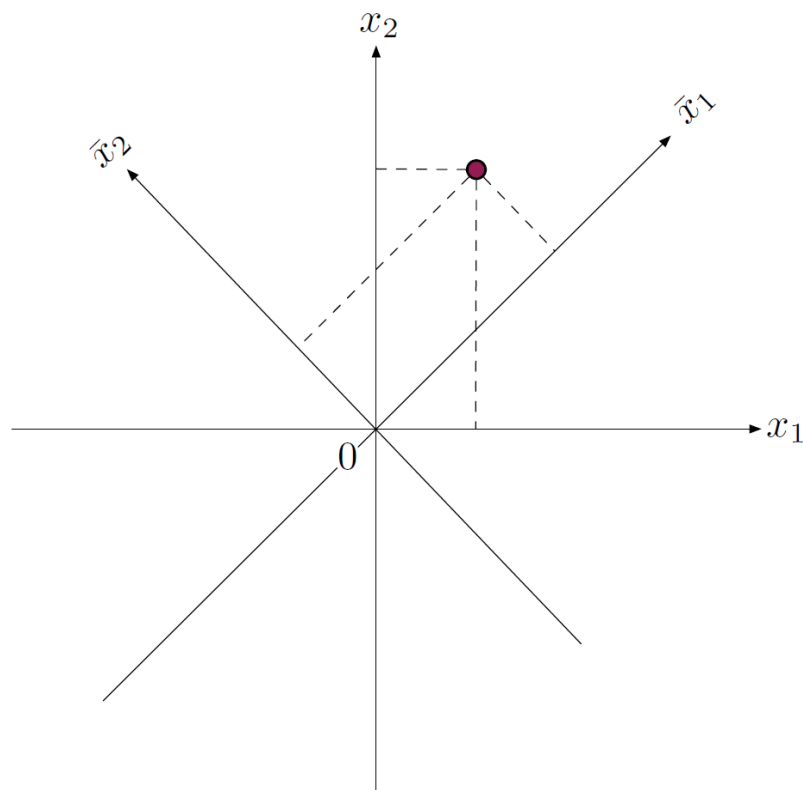
This is equivalent to having no RHP open-loop poles and no pole-zero cancellations in RHP.

Coordinate Transformations

Now that we have seen that a given transfer function can have many different state-space realizations, we would like a systematic procedure of generating such realizations, preferably with favorable properties (like controllability).

One such procedure is by means of *coordinate transformations*.

Coordinate Transformations



$$x \longmapsto \bar{x} = Tx,$$

$$T \in \mathbb{R}^{n \times n} \text{ nonsingular}$$

$$x = T^{-1}\bar{x}$$

(go back and forth between the coordinate systems)

Coordinate Transformations

For example,

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mapsto \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ x_1 - x_2 \end{pmatrix}$$

This can be represented as

$$\bar{x} = Tx, \quad \text{where } T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The transformation is invertible: $\det T = -2$, and

$$T^{-1} = \frac{1}{\det T} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

Or we can see this directly:

$$\bar{x}_1 + \bar{x}_2 = 2x_1; \quad \bar{x}_1 - \bar{x}_2 = 2x_2$$

Coordinate Transformations and State-Space Models

Consider a state-space model

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

and a change of coordinates $\bar{x} = Tx$ (T invertible).

What does the system look like in the new coordinates?

$$\dot{\bar{x}} = T\dot{x} = T\dot{x} \quad (\text{linearity of derivative})$$

$$= T(Ax + Bu)$$

$$= T(AT^{-1}\bar{x} + Bu) \quad (x = T^{-1}\bar{x})$$

$$= \underbrace{TAT^{-1}}_{\bar{A}}\bar{x} + \underbrace{TB}_{\bar{B}}u$$

$$y = Cx$$

$$= \underbrace{CT^{-1}}_{\bar{C}}\bar{x}$$

Coordinate Transformations and State-Space Models

$$\begin{array}{ccc} \dot{x} = Ax + Bu & \xrightarrow{T} & \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ y = Cx & & y = \bar{C}\bar{x} \end{array}$$

where

$$\bar{A} = TAT^{-1}, \quad \bar{B} = TB, \quad \bar{C} = CT^{-1}$$

What happens to

- ▶ the transfer function?
- ▶ the controllability matrix?

Coordinate Transformations and State-Space Models

$$\begin{array}{ccc} \dot{x} = Ax + Bu & \xrightarrow{T} & \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ y = Cx & & y = \bar{C}\bar{x} \end{array}$$

$$\text{where } \bar{A} = TAT^{-1}, \quad \bar{B} = TB, \quad \bar{C} = CT^{-1}$$

Claim: The transfer function doesn't change.

Proof:

$$\begin{aligned} \bar{G}(s) &= \bar{C}(Is - \bar{A})^{-1}\bar{B} \\ &= (CT^{-1})(Is - TAT^{-1})^{-1}(TB) \\ &= CT^{-1}(TIT^{-1}s - TAT^{-1})^{-1}TB \\ &= CT^{-1}[T(Is - A)T^{-1}]^{-1}TB \\ &= C \underbrace{T^{-1}T}_I (Is - A)^{-1} \underbrace{T^{-1}T}_I B \\ &= C(Is - A)^{-1}B \equiv G(s) \end{aligned}$$

Coordinate Transformations and State-Space Models

$$\begin{array}{ccc} \dot{x} = Ax + Bu & \xrightarrow{T} & \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ y = Cx & & y = \bar{C}\bar{x} \end{array}$$

$$\text{where } \bar{A} = TAT^{-1}, \quad \bar{B} = TB, \quad \bar{C} = CT^{-1}$$

The transfer function doesn't change.

In fact:

- ▶ open-loop poles don't change
- ▶ characteristic polynomial doesn't change:

$$\begin{aligned} \det(Is - \bar{A}) &= \det(Is - TAT^{-1}) \\ &= \det [T(Is - A)^{-1}T^{-1}] \\ &= \det T \cdot \det(Is - A)^{-1} \cdot \det T^{-1} \\ &= \det(Is - A)^{-1} \end{aligned}$$

Coordinate Transformations and State-Space Models

$$\begin{array}{ccc} \dot{x} = Ax + Bu & \xrightarrow{T} & \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ y = Cx & & y = \bar{C}\bar{x} \end{array}$$

$$\text{where } \bar{A} = TAT^{-1}, \quad \bar{B} = TB, \quad \bar{C} = CT^{-1}$$

Claim: Controllability doesn't change.

Proof: For any $k = 0, 1, \dots$,

$$\bar{A}^k \bar{B} = (TAT^{-1})^k TB = TA^k T^{-1} TB = TA^k B \quad (\text{by induction})$$

$$\begin{aligned} \text{Therefore, } \mathcal{C}(\bar{A}, \bar{B}) &= [TB \mid TAB \mid \dots \mid TA^{n-1}B] \\ &= T[B \mid AB \mid \dots \mid A^{n-1}B] \\ &= T\mathcal{C}(A, B) \end{aligned}$$

Since $\det T \neq 0$, $\det \mathcal{C}(\bar{A}, \bar{B}) \neq 0$ if and only if $\det \mathcal{C}(A, B) \neq 0$.

Thus, the new system is controllable if and only if the old one is.

Coordinate Transformations and State-Space Models

$$\begin{array}{ccc} \dot{x} = Ax + Bu & \xrightarrow{T} & \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ y = Cx & & y = \bar{C}\bar{x} \end{array}$$

$$\text{where } \bar{A} = TAT^{-1}, \quad \bar{B} = TB, \quad \bar{C} = CT^{-1}$$

Note: The *controllability matrix* does change:

$$\underbrace{\mathcal{C}(\bar{A}, \bar{B})}_{\text{new}} = \underbrace{T}_{\text{coord. trans.}} \underbrace{\mathcal{C}(A, B)}_{\text{old}}$$

$$\Updownarrow$$

$$T = \mathcal{C}(\bar{A}, \bar{B}) [\mathcal{C}(A, B)]^{-1}$$

This is a recipe for going from one *controllable* realization of a given t.f. to another.

CCF is the most convenient controllable realization of a given t.f., so we want to *convert a given controllable system to CCF* (useful for control design).

Example: Converting a Controllable System to CCF

Consider $A = \begin{pmatrix} -15 & 8 \\ -15 & 7 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Convert to CCF if possible.

Step 1: check for controllability.

$$AB = \begin{pmatrix} -15 & 8 \\ -15 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -7 \\ -8 \end{pmatrix} \implies \mathcal{C} = \begin{pmatrix} 1 & -7 \\ 1 & -8 \end{pmatrix}$$

$\det \mathcal{C} = -1$ — controllable

Example: Converting a Controllable System to CCF

Step 2: Determine desired $\mathcal{C}(\bar{A}, \bar{B})$.

We need to figure out $\bar{A}$ and $\bar{B}$.

For CCF, we must have

$$\bar{A} = \begin{pmatrix} 0 & 1 \\ -a_2 & -a_1 \end{pmatrix}, \quad \bar{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so we need to find the coefficients a_1, a_2 .

Recall: the characteristic polynomial does not change:

$$\begin{aligned} \det(Is - A) &= \det(Is - \bar{A}) \\ \det \begin{pmatrix} s + 15 & -8 \\ 15 & s - 7 \end{pmatrix} &= \det \begin{pmatrix} s & -1 \\ a_2 & s + a_1 \end{pmatrix} \\ (s + 15)(s - 7) + 120 &= s(s + a_1) + a_2 \\ s^2 + 8s + 15 &= s^2 + a_1s + a_2 \end{aligned}$$

Example: Converting a Controllable System to CCF

Step 2: Determine desired $\mathcal{C}(\bar{A}, \bar{B})$.

We need to figure out $\bar{A}$ and $\bar{B}$.

For CCF, we must have

$$\bar{A} = \begin{pmatrix} 0 & 1 \\ -a_2 & -a_1 \end{pmatrix}, \quad \bar{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

We have just computed

$$\bar{A} = \begin{pmatrix} 0 & 1 \\ -15 & -8 \end{pmatrix}, \quad \bar{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Therefore, the new controllability matrix should be

$$\mathcal{C}(\bar{A}, \bar{B}) = [\bar{B} \mid \bar{A}\bar{B}] = \begin{pmatrix} 0 & 1 \\ 1 & -8 \end{pmatrix}$$

Example: Converting a Controllable System to CCF

Step 3: Compute T .

Recall: $T = \mathcal{C}(\bar{A}, \bar{B}) \cdot [\mathcal{C}(A, B)]^{-1}$

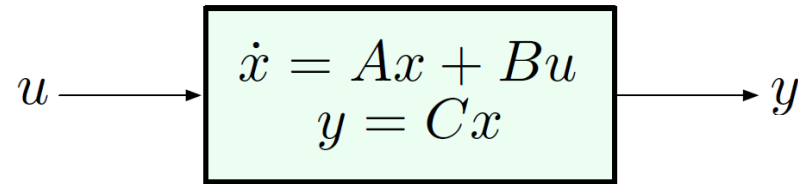
$$\mathcal{C}(A, B) = \begin{pmatrix} 1 & -7 \\ 1 & -8 \end{pmatrix}$$

$$\begin{aligned} [\mathcal{C}(A, B)]^{-1} &= \begin{pmatrix} 1 & -7 \\ 1 & -8 \end{pmatrix}^{-1} \\ &= \frac{1}{-1} \begin{pmatrix} -8 & 7 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 8 & -7 \\ 1 & -1 \end{pmatrix} \end{aligned}$$

$$\mathcal{C}(\bar{A}, \bar{B}) = \begin{pmatrix} 0 & 1 \\ 1 & -8 \end{pmatrix}$$

$$\begin{aligned} T &= \begin{pmatrix} 0 & 1 \\ 1 & -8 \end{pmatrix} \begin{pmatrix} 8 & -7 \\ 1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

State-Space Realizations

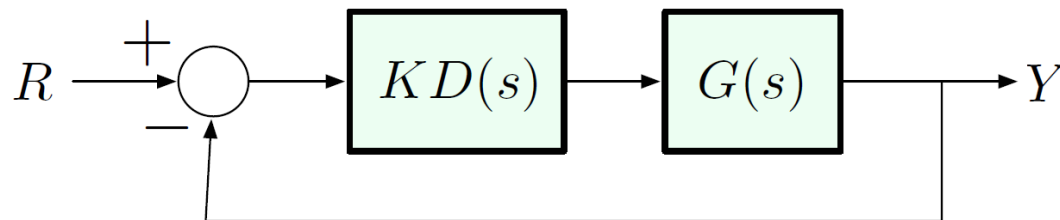


$$G(s) = C(Is - A)^{-1}B$$

Open-loop poles are the eigenvalues of A :

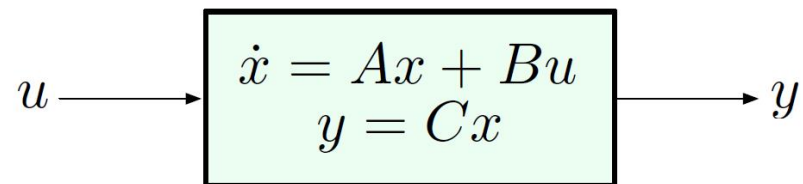
$$\det(Is - A) = 0$$

Then we add a controller to move the poles to desired locations:



Goal: Pole Placement by State Feedback

Consider a SISO system in state-space form:



Today, our goal is to establish the following fact:

If the above system is *controllable*, then we can assign arbitrary closed-loop poles by means of a **state feedback law**

$$\begin{aligned} u &= -Kx = -\begin{pmatrix} k_1 & k_2 & \dots & k_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \\ &= -(k_1 x_1 + \dots + k_n x_n), \end{aligned}$$

where K is a $1 \times n$ matrix of feedback gains.

Review: Controllability

Consider a single-input system ($u \in \mathbb{R}$):

$$\dot{x} = Ax + Bu, \quad y = Cx \quad x \in \mathbb{R}^n$$

The **Controllability Matrix** is defined as

$$\mathcal{C}(A, B) = [B \mid AB \mid A^2B \mid \dots \mid A^{n-1}B]$$

We say that the above system is **controllable** if its controllability matrix $\mathcal{C}(A, B)$ is *invertible*.

- ▶ As we will see today, if the system is controllable, then we may assign arbitrary closed-loop poles by *state feedback* of the form $u = -Kx$.
- ▶ Whether or not the system is controllable depends on its state-space realization.

Controller Canonical Form

A single-input state-space model

$$\dot{x} = Ax + Bu, \quad y = Cx$$

is said to be in **Controller Canonical Form** (CCF) if the matrices A, B are of the form

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ * & * & * & \dots & * & * \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

A system in CCF is *always controllable!!*

(The proof of this for $n > 2$ uses the Jordan canonical form, we will not worry about this.)

Coordinate Transformations

- ▶ We will see that state feedback design is particularly easy when the system is in CCF.
- ▶ Hence, we need a way of constructing a CCF state-space realization of a given controllable system.
- ▶ We will do this by suitably changing the coordinate system for the state vector.

Coordinate Transformations and State-Space Models

$$\begin{array}{ccc} \dot{x} = Ax + Bu & \xrightarrow{T} & \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ y = Cx & & y = \bar{C}\bar{x} \end{array}$$

where $\bar{A} = TAT^{-1}$, $\bar{B} = TB$, $\bar{C} = CT^{-1}$

- ▶ The transfer function does not change.
- ▶ The controllability matrix is transformed:

$$\mathcal{C}(\bar{A}, \bar{B}) = T\mathcal{C}(A, B).$$

- ▶ The transformed system is controllable if and only if the original one is.
- ▶ If the original system is controllable, then

$$T = \mathcal{C}(\bar{A}, \bar{B}) [\mathcal{C}(A, B)]^{-1}.$$

This gives us a way of systematically passing to CCF.

Finally, Pole Placement via State Feedback

Consider a state-space model

$$\dot{x} = Ax + Bu, \quad x \in \mathbb{R}^n, u \in \mathbb{R}$$

$$y = x$$

Let's introduce a *state feedback law*

$$u = -Ky \equiv -Kx$$

$$= - \begin{pmatrix} k_1 & k_2 & \dots & k_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = -(k_1 x_1 + \dots + k_n x_n)$$

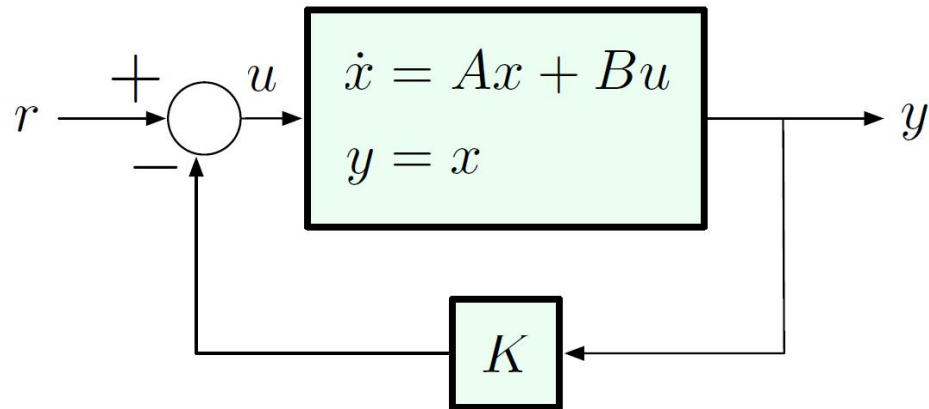
Closed-loop system:

$$\dot{x} = Ax - BKx = (A - BK)x$$

$$y = x$$

Pole Placement via State Feedback

Let's also add a reference input:



$$\dot{x} = Ax + B(-Kx + r) = (A - BK)x + Br, \quad y = x$$

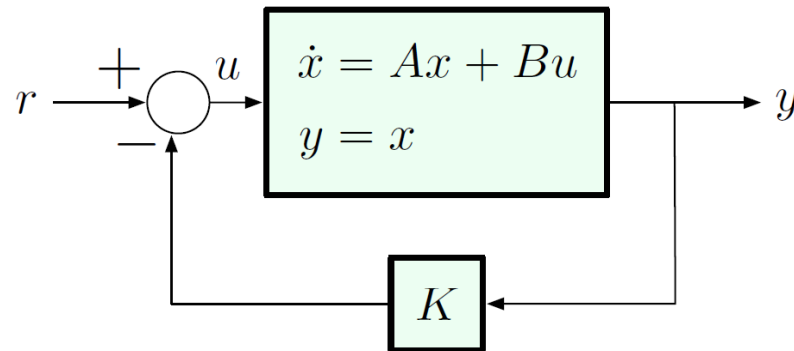
Take the Laplace transform:

$$sX(s) = (A - BK)X(s) + BR(s), \quad Y(s) = X(s)$$

$$Y(s) = \underbrace{C(Is - A + BK)^{-1}B}_{G} R(s)$$

Closed-loop poles are the eigenvalues of $A - BK$!!

Pole Placement via State Feedback



assigning closed-loop poles = assigning eigenvalues of $A - BK$

Now we will see that this is particularly straightforward if the (A, B) system is in CCF.

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \dots & -a_2 & -a_1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

The Beauty of CCF

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \dots & -a_2 & -a_1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

Claim.

$$\det(Is - A) = s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n$$

— the last row of the A matrix in CCF consists of the coefficients of the characteristic polynomial, in reverse order, with “ $-$ ” signs.

... And, Back to Pole Placement

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \dots & -a_2 & -a_1 \end{pmatrix}$$

$$BK = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} (k_1 \quad k_2 \quad \dots \quad k_n) = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 0 \\ k_1 & k_2 & k_3 & \dots & k_{n-1} & k_n \end{pmatrix}$$

$$A - BK = - \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ a_n + k_1 & a_{n-1} + k_2 & a_{n-2} + k_3 & \dots & a_2 + k_{n-1} & a_1 + k_n \end{pmatrix}$$

— still in CCF!!

Pole Placement in CCF

$$\dot{x} = (A - BK)x + Br, \quad y = Cx$$

$$A - BK = - \begin{pmatrix} 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 1 \\ a_n + k_1 & a_{n-1} + k_2 & \dots & a_2 + k_{n-1} & a_1 + k_n \end{pmatrix}$$

Closed-loop poles are the roots of the characteristic polynomial

$$\begin{aligned} \det(Is - A + BK) \\ = s^n + (a_2 + k_{n-1})s^{n-1} + \dots + (a_{n-1} + k_2)s + (a_n + k_1) \end{aligned}$$

Key observation: When the system is in CCF, each control gain affects only *one* of the coefficients of the characteristic polynomial, and these coefficients can be assigned arbitrarily by a suitable choice of $k_1, \dots, k_n$.

Hence the name **Controller Canonical Form** — convenient for control design.

Pole Placement by State Feedback

General procedure for any *controllable* system:

1. Convert to CCF using a suitable invertible coordinate transformation T (such a transformation exists by controllability).
2. Solve the pole placement problem in the new coordinates.
3. Convert back to original coordinates.

Example

Given $\dot{x} = Ax + Bu$

$$A = \begin{pmatrix} -15 & 8 \\ -7 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Goal: apply state feedback to place closed-loop poles at $-10 \pm j$.

Step 1: convert to CCF — already did this

$$T = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \quad \longrightarrow \quad \bar{A} = \begin{pmatrix} 0 & 1 \\ -15 & -8 \end{pmatrix}, \quad \bar{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Example

Step 2: find $u = -\bar{K}\bar{x}$ to place closed-loop poles at $-10 \pm j$.

Desired characteristic polynomial:

$$(s + 10 + j)(s + 10 - j) = (s + 10)^2 + 1 = s^2 + 20s + 101$$

Thus, the closed-loop system matrix should be

$$\bar{A} - \bar{B}\bar{K} = \begin{pmatrix} 0 & 1 \\ -101 & -20 \end{pmatrix}$$

On the other hand, we know

$$\bar{A} - \bar{B}\bar{K} = \begin{pmatrix} 0 & 1 \\ -(15 + \bar{k}_1) & -(8 + \bar{k}_2) \end{pmatrix} \implies \bar{k}_1 = 86, \bar{k}_2 = 12$$

This gives the control law

$$u = -\bar{K}\bar{x} = - \begin{pmatrix} 86 & 12 \end{pmatrix} \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \end{pmatrix}$$

Example

Step 3: convert back to the old coordinates.

$$\begin{aligned} u &= -\bar{K}\bar{x} \\ &= -\underbrace{\bar{K}T}_K x \end{aligned}$$

— therefore,

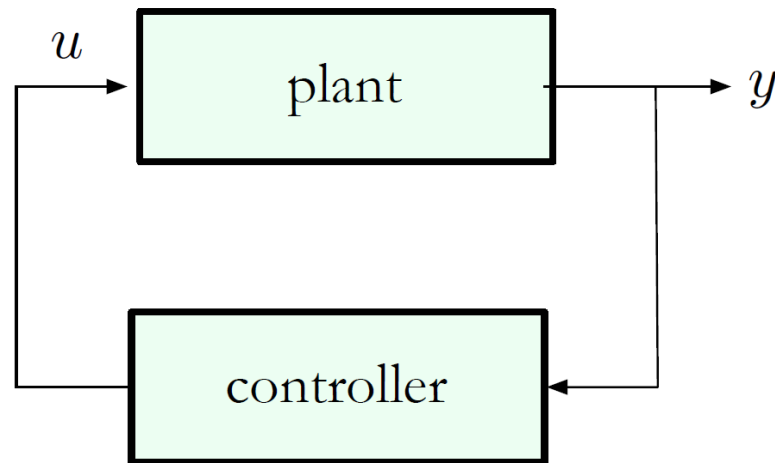
$$\begin{aligned} K &= \bar{K}T \\ &= (86 \quad 12) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \\ &= (86 \quad -74) \end{aligned}$$

The desired state feedback law is

$$u = (-86 \quad 74) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

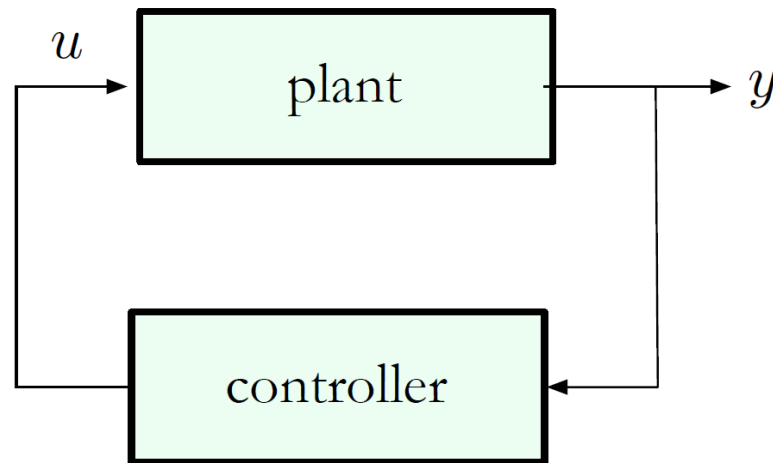
Is Full State Feedback Always Available?

In a typical system, measurements are provided by sensors:



Is Full State Feedback Always Available?

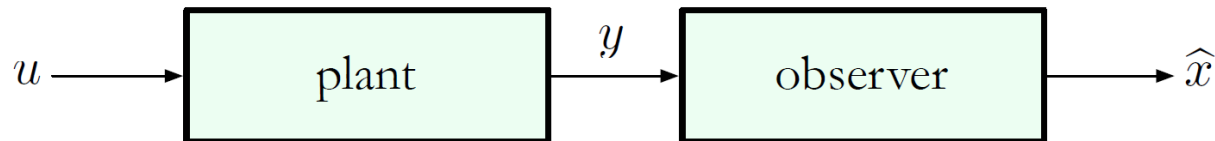
In a typical system, measurements are provided by sensors:



Full state feedback $u = -Kx$ is *not implementable!!*

State Estimation Using an Observer

When full state feedback is unavailable, the **observer** is used to **estimate** the state x :

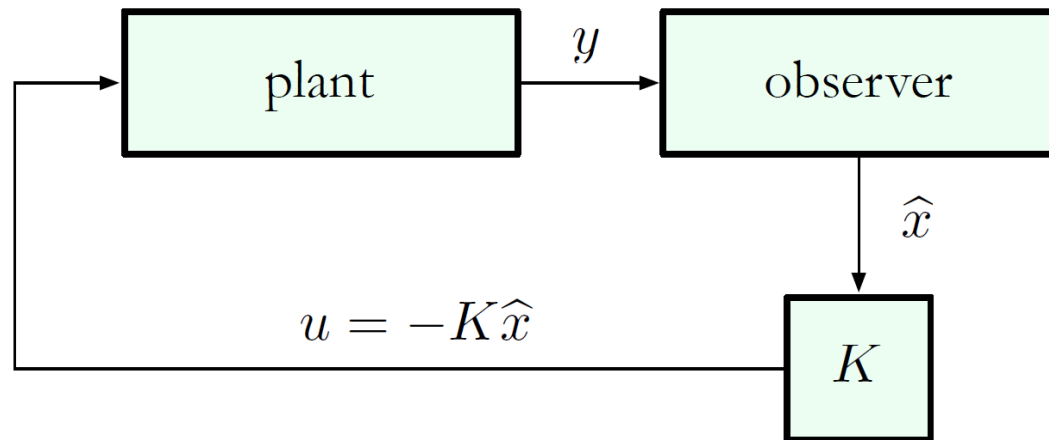


State Estimation Using an Observer

The idea is to design the observer in such a way that the state estimate $\hat{x}$ is *asymptotically accurate*:

$$\|\hat{x}(t) - x(t)\| = \sqrt{\sum_{i=1}^n (\hat{x}_i(t) - x_i(t))^2} \xrightarrow{t \rightarrow \infty} 0$$

If we are successful, then we can try **estimated state feedback**:



A New Concept: Observability

- ▶ Before, we saw that closed-loop poles can be assigned arbitrarily by full state feedback when the plant is controllable.
- ▶ Now, we will see that asymptotically accurate state estimation will be possible when the system is observable.
- ▶ Observability is a system property which is dual to controllability.

Observability

Consider a single-output system ($y \in \mathbb{R}$):

$$\dot{x} = Ax + Bu, \quad y = Cx \quad x \in \mathbb{R}^n$$

The **Observability Matrix** is defined as

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

- recall that C is $1 \times n$ and A is $n \times n$, so $\mathcal{O}(A, C)$ is $n \times n$;
- the controllability matrix only involves A and C , not B

We say that the above system is **observable** if its controllability matrix $\mathcal{O}(A, C)$ is *invertible*.

(This definition is only true for the single-output case; the multiple-output case involves the *rank* of $\mathcal{O}(A, C)$.)

Example: Computing $\mathcal{O}(A, C)$

$$\text{Let } A = \begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix}, \quad C = (0 \quad 1)$$

Here, $n = 2$, $C \in \mathbb{R}^{1 \times 2}$, $A \in \mathbb{R}^{2 \times 2} \implies \mathcal{O}(A, C) \in \mathbb{R}^{2 \times 2}$.

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ CA \end{bmatrix}$$

$$\text{where } CA = (0 \quad 1) \begin{pmatrix} 0 & -6 \\ 1 & -5 \end{pmatrix} = (1 \quad -5)$$

$$\therefore \mathcal{O}(A, C) = \begin{pmatrix} 0 & 1 \\ 1 & -5 \end{pmatrix}$$

$$\det \mathcal{O}(A, C) = -1 \implies \text{the system is observable}$$

— recall: this system is in **Observer Canonical Form (OCF)** ...

Observer Canonical Form

A single-output state-space model

$$\dot{x} = Ax + Bu, \quad y = Cx$$

is said to be in **Observer Canonical Form (OCF)** if the matrices A, C are of the form

$$A = \begin{pmatrix} 0 & 0 & \dots & 0 & 0 & * \\ 1 & 0 & \dots & 0 & 0 & * \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 & * \\ 0 & 0 & \dots & 0 & 1 & * \end{pmatrix}, \quad C = (0 \quad 0 \quad \dots \quad 0 \quad 1)$$

Fact: A system in OCF is *always observable!!*

(The proof of this for $n > 2$ uses the Jordan canonical form, we will not worry about this.)

Coordinate Transformations and Observability

Just like controllability, observability is preserved under invertible coordinate transformations.

$$\begin{array}{ccc} \dot{x} = Ax + Bu & \xrightarrow{T} & \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ y = Cx & & y = \bar{C}\bar{x} \end{array}$$

$$\text{where } \bar{A} = TAT^{-1}, \quad \bar{B} = TB, \quad \bar{C} = CT^{-1}$$

$$\begin{aligned} \mathcal{O}(\bar{A}, \bar{C}) &= \begin{pmatrix} \bar{C} \\ \bar{C}\bar{A} \\ \vdots \\ \bar{C}\bar{A}^{n-1} \end{pmatrix} = \begin{pmatrix} CT^{-1} \\ CT^{-1}TAT^{-1} \\ \vdots \\ CT^{-1}TA^{n-1}T^{-1} \end{pmatrix} \\ &= \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix} T^{-1} = \mathcal{O}(A, C)T^{-1} \end{aligned}$$

Coordinate Transformations and Observability

Just like controllability, observability is preserved under invertible coordinate transformations:

$$\begin{array}{ccc} \dot{x} = Ax + Bu & \xrightarrow{T} & \dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}u \\ y = Cx & & y = \bar{C}\bar{x} \end{array}$$

$$\text{where } \bar{A} = TAT^{-1}, \quad \bar{B} = TB, \quad \bar{C} = CT^{-1}$$

If the original system is controllable, then

$$\begin{array}{c} \underbrace{T [\mathcal{O}(A, C)]^{-1}}_{\text{old}} = \underbrace{\mathcal{O}(\bar{A}, \bar{C})^{-1}}_{\text{new}} \\ \Updownarrow \\ T = \underbrace{\mathcal{O}(\bar{A}, \bar{C})^{-1}}_{\text{new}} \underbrace{[\mathcal{O}(A, C)]}_{\text{old}} \end{array}$$

Observability and State Estimation

As we will show next:

If the system is observable, then there exists an observer (state estimator) that provides an asymptotically convergent estimate $\hat{x}$ of the state x based on the observed output y .



The particular type of observer we will construct is called the **Luenberger observer** after David G. Luenberger, who developed this idea in his 1963 Ph.D. dissertation.

David Luenberger is a Professor at Stanford University.

The Luenberger Observer

Consider a state-space model

$$\begin{aligned}\dot{x} &= Ax && \text{(for now, assume } u = 0) \\ y &= Cx\end{aligned}$$

We wish to estimate the state x based on the output y .

Consider feeding the output y as input to the following system with state $\hat{x}$:

$$\dot{\hat{x}} = (A - LC)\hat{x} + Ly.$$

Assumption: The output injection matrix L is chosen in such a way that the matrix $A - LC$ is Hurwitz (i.e., all of its eigenvalues lie in LHP).

At this point, we do not assume anything about observability.

The Luenberger Observer

$$\text{System:} \quad \dot{x} = Ax$$

$$y = Cx$$

$$\text{Observer:} \quad \dot{\hat{x}} = (A - LC)\hat{x} + Ly.$$

What happens to **state estimation error** $e = x - \hat{x}$ as $t \rightarrow \infty$?

$$\begin{aligned} \dot{e} &= \dot{x} - \dot{\hat{x}} \\ &= Ax - [(A - LC)\hat{x} + LCx] \\ &= (A - LC)x - (A - LC)\hat{x} \\ &= (A - LC)e \end{aligned}$$

Does $e(t)$ converge to zero in some sense?

Linear ODEs and Eigenvalues: A Digression

$$\dot{v} = Fv, \quad v \in \mathbb{R}^n, \quad F \in \mathbb{R}^{n \times n}$$

Let $\lambda_1, \dots, \lambda_n$ be eigenvalues of F , i.e., roots of $\det(Is - F) = 0$.

Then there exists an invertible matrix $T \in \mathbb{R}^{n \times n}$, such that

$$F = T^{-1} \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} T$$

Consider the change of coordinates $\bar{v} = Tv$. Then

$$\dot{\bar{v}} = T F T^{-1} \bar{v} = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} \bar{v}$$

Linear ODEs: A Digression

$$\dot{\bar{v}} = T F T^{-1} \bar{v} = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} \bar{v}, \quad (\lambda_1, \dots, \lambda_n) = \text{eig}(F)$$

$\Updownarrow$

$$\dot{\bar{v}}_i = \lambda_i \bar{v}_i, \quad i = 1, 2, \dots, n$$

This system of n 1st-order ODEs has the solution

$$\bar{v}_i(t) = \bar{v}_i(0) e^{\lambda_i t}, \quad i = 1, 2, \dots, n$$

If all λ_i 's have negative real parts, then

$$\begin{aligned} \|v(t)\|^2 &= v(t)^T v(t) = \bar{v}(t)^T \bar{v}(t) \\ &\leq C e^{-2\sigma_{\min} t}, \end{aligned} \quad \text{where } \sigma_{\min} = \min_{1 \leq i \leq n} |\text{Re}(\lambda_i)|$$

The Luenberger Observer

$$\text{System:} \quad \dot{x} = Ax$$

$$y = Cx$$

$$\text{Observer:} \quad \dot{\hat{x}} = (A - LC)\hat{x} + Ly$$

$$\text{Error:} \quad \dot{e} = (A - LC)e$$

Recall our assumption that $A - LC$ is Hurwitz (all eigenvalues are in LHP). This implies that

$$\|x(t) - \hat{x}(t)\|^2 = \|e(t)\|^2 = \sum_{i=1}^n |e_i(t)|^2 \xrightarrow{t \rightarrow \infty} 0$$

at an exponential rate, determined by the eigenvalues of $A - LC$.

For fast convergence, want eigenvalues of $A - LC$ far into LHP!!

The Luenberger Observer

$$\text{System:} \quad \dot{x} = Ax$$

$$y = Cx$$

$$\text{Observer:} \quad \dot{\hat{x}} = (A - LC)\hat{x} + Ly$$

$$\text{Error:} \quad \dot{e} = (A - LC)e$$

Observer transfer function:

$$s\hat{X}(s) = (A - LC)\hat{X}(s) + LY(s)$$

$$(Is - A + LC)\hat{X}(s) = LY(s)$$

$$\hat{X}(s) = (Is - A + LC)^{-1}LY(s).$$

The eigenvalues of $A - LC$ are the **observer poles**. We want these poles to be *stable* and *fast*.

Observability and Estimation Error

Fact: If the system

$$\dot{x} = Ax, \quad y = Cx$$

is observable, then we can arbitrarily assign eigenvalues of $A - LC$ by a suitable choice of the output injection matrix L .

This is similar to the fact that controllability implies arbitrary closed-loop pole placement by state feedback.

In fact, these two facts are closely related because CCF is dual to OCF.

Observer Pole Placement in OCF

Consider a single-output system in OCF:

$$\dot{x} = Ax$$

$$y = Cx, \quad y \in \mathbb{R}$$

$$\text{where } A = \begin{pmatrix} 0 & 0 & \dots & 0 & 0 & -a_n \\ 1 & 0 & \dots & 0 & 0 & -a_{n-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 & -a_2 \\ 0 & 0 & \dots & 0 & 1 & -a_1 \end{pmatrix}, \quad C = (0 \quad 0 \quad \dots \quad 0 \quad 1)$$

Note that A^T has the form of a CCF system matrix, thus:

$$\begin{aligned} \det(Is - A) &= \det((Is - A)^T) = \det(Is - A^T) \\ &= s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n \end{aligned}$$

Now Let's Add an Observer

$$A = \begin{pmatrix} 0 & 0 & \dots & 0 & -a_n \\ 0 & 1 & \dots & 0 & -a_{n-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & -a_2 \\ 0 & 0 & \dots & 1 & -a_1 \end{pmatrix}$$

$$LC = \begin{pmatrix} \ell_1 \\ \ell_2 \\ \vdots \\ \ell_n \end{pmatrix} (0 \quad 0 \quad \dots \quad 1) = \begin{pmatrix} 0 & 0 & \dots & 0 & \ell_1 \\ 0 & 0 & \dots & 0 & \ell_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & \ell_{n-1} \\ 0 & 0 & \dots & 0 & \ell_n \end{pmatrix}$$

$$A - LC = \begin{pmatrix} 0 & 0 & \dots & 0 & -(a_n + \ell_1) \\ 0 & 1 & \dots & 0 & -(a_{n-1} + \ell_2) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & -(a_2 + \ell_{n-1}) \\ 0 & 0 & \dots & 1 & -(a_1 + \ell_n) \end{pmatrix}$$

— still in OCF!!

Observer Pole Placement in OCF

$$\dot{x} = Ax, \quad y = Cx, \quad \dot{\hat{x}} = (A - LC)\hat{x} + Ly$$

$$A - LC = \begin{pmatrix} 0 & 0 & \dots & 0 & -(a_n + \ell_1) \\ 1 & 0 & \dots & 0 & -(a_{n-1} + \ell_2) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & -(a_2 + \ell_{n-1}) \\ 0 & 0 & \dots & 1 & -(a_1 + \ell_n) \end{pmatrix}$$

Eigenvalues of $A - LC$ are the roots of the characteristic polynomial

$$\begin{aligned} \det(Is - A + LC) \\ = s^n + (a_1 + \ell_n)s^{n-1} + \dots + (a_{n-1} + \ell_2)s + (a_n + \ell_1) \end{aligned}$$

Key observation: In OCF, each observer gain affects only *one* of the coefficients of the characteristic polynomial, which can be assigned arbitrarily by a suitable choice of $\ell_1, \dots, \ell_n$.

Hence the name **Observer Canonical Form** — convenient for observer design.

Observer Pole Placement

General procedure for any *observable* system:

1. Convert to OCF: $T = \underbrace{\mathcal{O}(\bar{A}, \bar{C})^{-1}}_{\text{new}} \underbrace{[\mathcal{O}(A, C)]}_{\text{old}}$
2. Find $\bar{L}$, such that $\bar{A} - \bar{L}\bar{C}$ has desired eigenvalues.
3. Convert back to original coordinates: $L = T^{-1}\bar{L}$.

The resulting observer is

$$\dot{\hat{x}} = (A - T^{-1}\bar{L}C)\hat{x} + T^{-1}\bar{L}y$$

In fact, this procedure is not necessary because of duality between controllability and observability!!

Controllability–Observability Duality

Claim: The system

$$\dot{x} = Ax, \quad y = Cx$$

is observable if and only if the system

$$\dot{x} = A^T x + C^T u$$

is controllable.

Proof: $\mathcal{C}(A^T, C^T) = [C^T \mid A^T C^T \mid \dots \mid (A^T)^{n-1} C^T]$

$$= \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}^T = [\mathcal{O}(A, C)]^T$$

Thus, $\mathcal{O}(A, C)$ is nonsingular if and only if $\mathcal{C}(A^T, C^T)$ is.

Observer Pole Placement, O/C Duality Version

Given an **observable** pair (A, C) :

1. For $F = A^T$, $G = C^T$, consider the system $\dot{x} = Fx + Gu$ (this system is controllable).
2. Use our earlier procedure to find K , such that

$$F - GK = A^T - C^T K$$

has desired eigenvalues.

3. Then

$$\text{eig}(A^T - C^T K) = \text{eig}(A^T - C^T K)^T = \text{eig}(A - K^T C),$$

so $L = K^T$ is the desired output injection matrix.

Final answer: use the observer

$$\begin{aligned}\dot{\hat{x}} &= (A - LC)\hat{x} + Ly \\ &= (A - K^T C)\hat{x} + K^T y.\end{aligned}$$

Combining Full-State Feedback with an Observer

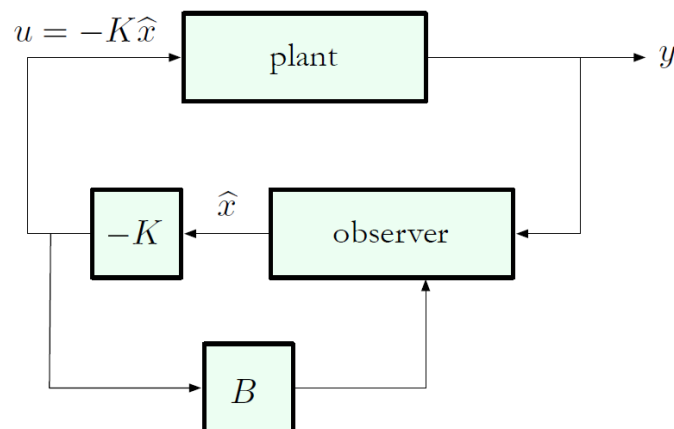
- ▶ So far, we have focused on autonomous systems ($u = 0$).
- ▶ What about nonzero inputs?

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

— assume (A, B) is controllable and (A, C) is observable.

- ▶ In the next lecture, we will learn how to use an observer together with estimated state feedback to (approximately) place closed-loop poles.



Combining Full-State Feedback with an Observer

- ▶ Consider

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

where (A, B) is controllable and (A, C) is observable.

- ▶ We know how to find K , such that $A - BK$ has desired eigenvalues (controller poles).
- ▶ Since we do not have access to x , we must design an observer. But this time, we need a slight modification because of the Bu term.

Observer in the Presence of Control Input

- ▶ Let's see what goes wrong when we use the old approach:

$$\dot{\hat{x}} = (A - LC)\hat{x} + Ly$$

- ▶ For the estimation error $e = x - \hat{x}$, we have

$$\begin{aligned}\dot{e} &= \dot{x} - \dot{\hat{x}} \\ &= Ax + Bu - [(A - LC)\hat{x} + LCx] \\ &= (A - LC)e + Bu \quad \text{-- not good}\end{aligned}$$

- ▶ **Idea:** since u is a signal we can access, let's use it as an input to the observer to cancel the Bu term from $\dot{x}$.
- ▶ Modified observer:

$$\begin{aligned}\dot{\hat{x}} &= (A - LC)\hat{x} + Ly + Bu \\ \dot{e} &= \dot{x} - \dot{\hat{x}} \\ &= Ax + Bu - [(A - LC)\hat{x} + LCx + Bu] \\ &= (A - LC)e \quad \text{regardless of } u\end{aligned}$$

Observer and Controller

$$\text{System: } \dot{x} = Ax + Bu$$

$$y = Cx$$

$$\text{Observer: } \dot{\hat{x}} = (A - LC)\hat{x} + Ly + Bu$$

$$\text{Error: } \dot{e} = (A - LC)e$$

- By observability, we can arbitrarily assign $\text{eig}(A - LC)$; these should be farther into LHP than desired controller poles.

$$\text{Controller: } u = -K\hat{x} \quad (\text{estimated state feedback})$$

- By controllability, we can arbitrarily assign $\text{eig}(A - BK)$.

Observer and Controller

$$\text{System: } \dot{x} = Ax + Bu$$

$$y = Cx$$

$$\text{Observer: } \dot{\hat{x}} = (A - LC)\hat{x} + Ly + Bu$$

$$\text{Controller: } u = -K\hat{x}$$

The overall observer-controller system is:

$$\dot{\hat{x}} = (A - LC)\hat{x} + Ly + B \underbrace{(-K\hat{x})}_{=u}$$

$$= (A - LC - BK)\hat{x} + Ly$$

$$u = -K\hat{x} \quad (\text{dynamic output feedback})$$

— this is a dynamical system with input y and output u

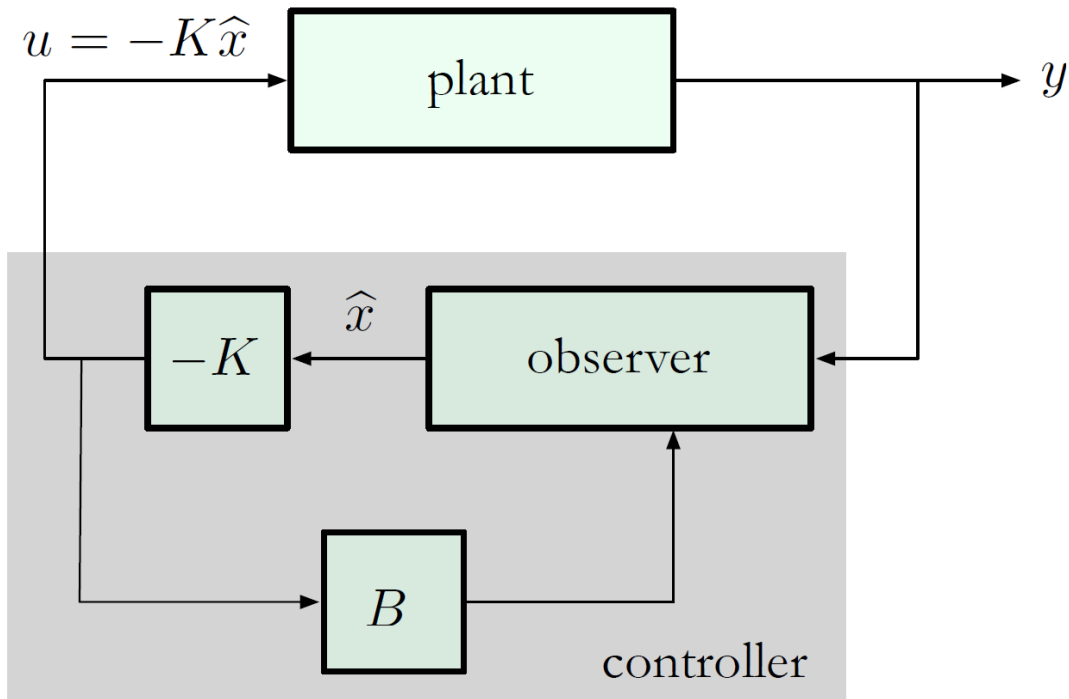
Dynamic Output Feedback

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

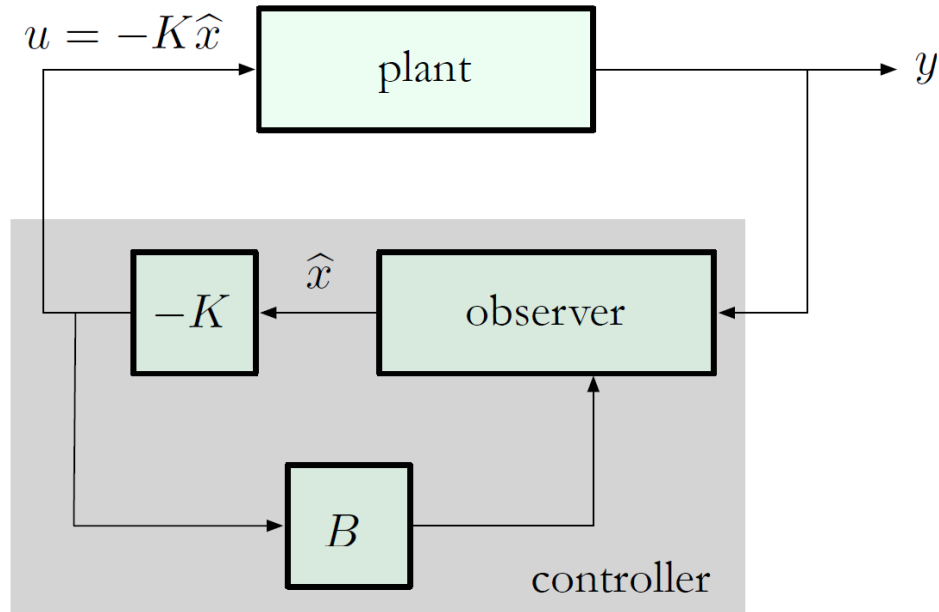
$$\dot{\hat{x}} = (A - LC - BK)\hat{x} + Ly$$

$$u = -K\hat{x}$$



Dynamic Output Feedback

$$\dot{\hat{x}} = (A - LC - BK)\hat{x} + Ly, \quad u = -K\hat{x}$$



Controller transfer function (from y to u):

$$s\hat{X} = (A - LC - BK)\hat{X} + LY, \quad U = -K\hat{X}$$

$$U = \underbrace{-K(Is - A + LC + BK)^{-1}LY}_{=D(s)}$$

Dynamic Output Feedback: Does It Work?

Summarizing:

- ▶ When $y = x$, full state feedback $u = -Kx$ achieves desired pole placement.
- ▶ How do we know that $u = -K\hat{x}$ achieves similar objectives?

Here is our overall closed-loop system:

$$\begin{aligned}\dot{x} &= Ax - BK\hat{x} \\ \dot{\hat{x}} &= (A - LC - BK)\hat{x} + LCx\end{aligned}$$

We can write it in block matrix form:

$$\begin{pmatrix} \dot{x} \\ \dot{\hat{x}} \end{pmatrix} = \begin{pmatrix} A & -BK \\ LC & A - LC - BK \end{pmatrix} \begin{pmatrix} x \\ \hat{x} \end{pmatrix}$$

How do we relate this to “nominal” behavior, $A - BK$?

Dynamic Output Feedback

$$\begin{pmatrix} \dot{x} \\ \dot{\hat{x}} \end{pmatrix} = \begin{pmatrix} A & -BK \\ LC & A - LC - BK \end{pmatrix} \begin{pmatrix} x \\ \hat{x} \end{pmatrix}$$

Let us transform to new coordinates:

$$\begin{pmatrix} x \\ \hat{x} \end{pmatrix} \mapsto \begin{pmatrix} x \\ e \end{pmatrix} = \begin{pmatrix} x \\ x - \hat{x} \end{pmatrix} = \underbrace{\begin{pmatrix} I & 0 \\ I & -I \end{pmatrix}}_T \begin{pmatrix} x \\ \hat{x} \end{pmatrix}$$

Two key observations:

- ▶ T is invertible, so the new representation is equivalent to the old one
- ▶ in the new coordinates, we have

$$\begin{aligned} \dot{x} &= Ax - BK\hat{x} \\ &= (A - BK)x + BK(x - \hat{x}) \\ &= (A - BK)x + BKe \\ \dot{e} &= (A - LC)e \end{aligned}$$

The Main Result: Separation Principle

So now we can write

$$\begin{pmatrix} \dot{x} \\ \dot{e} \end{pmatrix} = \underbrace{\begin{pmatrix} A - BK & BK \\ 0 & A - LC \end{pmatrix}}_{\text{upper triangular matrix}} \begin{pmatrix} x \\ e \end{pmatrix}$$

The closed-loop characteristic polynomial is

$$\begin{aligned} \det \begin{pmatrix} Is - A + BK & -BK \\ 0 & Is - A + LC \end{pmatrix} \\ = \det(Is - A + BK) \cdot \det(Is - A + LC) \end{aligned}$$

Separation principle. The closed-loop eigenvalues are:

$$\begin{aligned} &\{\text{controller poles (roots of } \det(Is - A + BK))\} \\ &\cup \{\text{observer poles (roots of } \det(Is - A + LC))\} \end{aligned}$$

— this holds only for linear systems!!

Separation Principle

Separation principle. The closed-loop eigenvalues are:

$$\begin{aligned} &\{\text{controller poles (roots of } \det(Is - A + BK))\} \\ &\cup \{\text{observer poles (roots of } \det(Is - A + LC))\} \end{aligned}$$

— this holds only for linear systems!!

Moral of the story:

- ▶ If we choose observer poles to be several times faster than the controller poles (e.g., 2–5 times), then the controller poles will be dominant.
- ▶ Dynamic output feedback gives essentially the same performance as (nonimplementable) full-state feedback — provided observer poles are far enough into LHP.
- ▶ Remember: the system must be **controllable** and **observable**!!